

Part A: Multiple Choice Write the letter of your choice in the space provided.

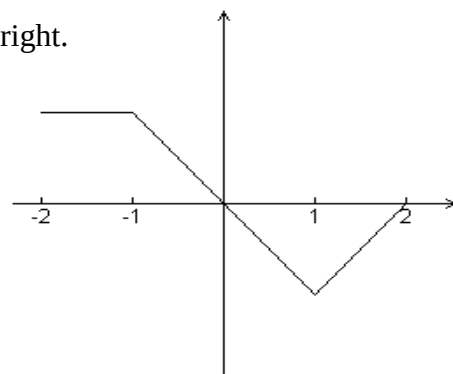
— 1) Let f be the function with derivative given by $f'(x) = x^2 - \frac{2}{x}$. On which of the following intervals is f decreasing?

- A) $(-\infty, -1]$ B) $(-\infty, 0)$ C) $[-1, 0)$ D) $(0, \sqrt[3]{2})$ E) $(\sqrt[3]{2}, \infty)$

— 2) If the line tangent to the graph of the function f at the point $(1, 7)$ passes through the point $(-2, -2)$, then $f'(1)$ is

- A) -5 B) 1 C) 3 D) 7 E) undefined

— 3) The graph of f' , the derivative of the function f , is shown at right. Which of the following statements is true about f ?



A) f is decreasing for $-1 < x < 1$.

B) f is increasing for $-2 < x < 0$.

C) f is increasing for $1 < x < 2$.

D) f has a local minimum at $x = 0$.

E) f is not differentiable at $x = -1$ and $x = 1$.

Graph of f'

— 4) Determine constants a and b such that the function $f(x) = x^3 + ax^2 + bx + c$ has a relative minimum at $x = 4$ and a point of inflection at $x = 1$.

- A) $a = 1, b = 3$ B) $a = -3, b = 3$ C) $a = -3, b = -24$
 D) $a = 3, b = 0$ E) $a = -6, b = 2$

— 5) Which of the following are true for $h(x) = x^4 - 4x$?

- I. h has a point of inflection at $x = 0$
 II. h has an absolute minimum of -3
 III. The second derivative test for local extrema is inconclusive

- A) I and II B) I and III C) II and III D) I only E) II only

Part B: Full Solutions

- Find the exact value of a such that the function $f(x) = \sqrt{x-2} - \frac{a}{x}$ has a point of inflection at $x=3$.
- Sketch $f(x) = x^{\frac{2}{3}}(8-x)^{\frac{1}{3}}$. Find the coordinates of all relative extrema and inflection Points.

$$f'(x) = \frac{16-3x}{3x^{\frac{1}{3}}(8-x)^{\frac{2}{3}}} \quad f''(x) = \frac{-128}{9x^{\frac{4}{3}}(8-x)^{\frac{5}{3}}}$$

- Perform a sketch for the following functions. Clearly indicate the results of each step.

$$\text{a) } f(x) = \frac{x^3 - 4x}{3x^2 + 9x} \quad , \quad f'(x) = \frac{x^2 + 6x + 4}{3(x+3)^2} \quad , \quad f''(x) = \frac{10}{3(x+3)^3}$$

$$\text{b) } f(x) = \frac{3x^2}{(x-3)^2} \quad , \quad f'(x) = \frac{-18x}{(x-3)^3} \quad , \quad f''(x) = \frac{18(2x+3)}{(x-3)^4}$$

$$\text{c) } f(x) = 4 - (x-3)^{\frac{2}{3}}$$

- Consider the function $f(x) = -x^3 + 6x^2 + 15x + 2$ on the closed interval $[-2, 2]$.
 - Find, and classify, all relative extrema.
 - Determine the absolute maximum of the function on the interval.
- Let $f(x) = x\left(4 + x^2 - \frac{x^4}{5}\right)$. Find the interval(s) on which f is increasing.
- Use the **SECOND DERIVATIVE TEST** to find the local extrema of the function $f(x) = 3x^4 + 8x^3 - 6x^2 - 24x + 3$.
- The function $f(x) = 2kx^3 + 3x^2 + px - 3$ has a local minimum at $x = -1$ and a point of inflection at $x = 1$. Determine the values of k and p .
- A function is defined by $f(x) = ax^3 + bx + c$.
 - Find the values of a , b , and c if $f(x)$ has a y-intercept at $(0, 2)$ and a local maximum at $(2, 6)$.
 - Explain how you know there must be local minimum.
- The slope of tangent to $g(x) = ax^3 + bx^2 + cx$ at its point of inflection $(1, 5)$ is 4. What are the values of a , b , and c .
- Sketch the graph of a rational function that satisfies all of the following conditions:

$$f''(x) < 0 \text{ when } x < -4 \quad f''(x) > 0 \text{ when } x > -4 \quad f'(x) < 0 \text{ for all } x$$

$$\lim_{x \rightarrow -4^+} f(x) = +\infty \quad \lim_{x \rightarrow -4^-} f(x) = -\infty \quad \lim_{x \rightarrow \pm\infty} f(x) = -2 \quad f(2) = 0$$